

Topic 6 of Personnel Economics: Monopsony and efficiency wages

Exercises

Exercise 6.1

Consider a perfectly competitive economy where workers can choose between two types of jobs: safe jobs and risky jobs. Safe jobs pay 50.000 euro per year and entail no risk whatsoever. When taking a risky job, workers have probability p of injury. Worker's utility function is:

$$U = w + g,$$

where w is the worker's income and g is the worker's health. When injured, $g = -100.000$. When not injured, $g = 0$.

a) Derive the level of the wage firms must offer to attract workers for a risky job.

Suppose $p=0.10$. At a yearly cost per worker of 7.500 euro, the probability of injury when having a risky job can be halved.

b) Will firms invest in reducing the probability of injury?

Suppose the government enforces firms to make the investment.

c) What happens with wages of risky jobs and prices of the products workers with risky jobs produce as a result of the government's safety regulation?

Suppose the government aims to maximize social welfare.

d) What could be the government's reason for enforcing firms to invest in safety?

Exercise 6.2 *(partly adapted from Peter Kuhn's teaching material)*

a) Derive the formula (17.2) on page 283 in the book expressing the wage as a function of the labor supply elasticity and the worker's productivity. If you struggle, review again how we did this in the main lecture.

b) *Imagine that cashiers at Alfredsons, a local supermarket chain, all process the same amount of orders in a typical week. Their hourly net productivity (after subtracting out all costs except for the cashiers' own labor) averages out to \$20. Using the 'optimal exploitation' formula, what is the profit-maximizing wage when the elasticity of cashier labor supply to the chain (η) equals 9? When $\eta = 4$? $\eta = 1.5$? $\eta = 1.0$? Explain in words why the profit-maximizing gap between productivity and wages ($Q-w$) is lower when the elasticity of labor supply is higher.*

Exercise 6.3

Suppose life consists of two periods of equal length. In the first period, people can either work or study. In the second period, all people work. People are assumed to care only about the present value of life-time consumption.

People differ in innate ability a . A person's wage depends on his ability and on his education. An individual with ability a who works in both periods earns income $w \cdot a$ per period. An individual with ability a who studies in the first period earns nothing in the first period, and income $w \cdot a + p \cdot a$ in the second period, where $p > 0$. Tuition and other direct cost of studying are equal to t . The per-period interest rate is equal to r . We also assume that $p > (1+r)w$.

a) Which people will decide to study in period 1? How does the interest rate affect the decision to study? How do the other variables affect the decision to study? Give interpretations for your results.

Suppose now that the government imposes a tax of rate τ on labor income. An economist warns that such a tax may weaken people's incentive to study.

b) Analyze whether the warning of the economist is justified. Explain your result.

Another economist argues that the distortionary effect of taxes on education decisions can be nullified by making tuition tax deductible (that is, if somebody has paid amount t tuition fee, then (s)he receives an amount of τt from the government in period 1).

c) Check whether the economist is right.

Exercise 6.4

A firm employs workers who have productivity 100 per year and are paid a yearly wage of 80. The firm is contemplating sending the workers to a specialised training, which costs the firm 50 per worker plus three months absence of each worker. The training increases each worker's yearly productivity by 10, starting the year following the training. Workers are identical, except for their year of retirement, which is denoted by T . The firm maximizes the present value of profits and has no competitors. The interest rate is r .

a) Which workers should the firm send to the specialised training? Interpret your result.

Suppose now that with probability π , a worker will decide to retire after $T - 10$ years rather than after T years. With the remaining probability, the worker will retire after T years.

b) Under which condition should the firm send a worker to the specialised training? Interpret your result.

The manager wants to reduce the risk of sending workers to the specialised training who know that they want to retire early. Simply asking the workers about their retirement plans could help, but the manager fears that not all of the workers will respond honestly as they seem to like to go to the training resort. The manager therefore thinks about structuring workers' compensation in such a way that they reveal their retirement plans.

c) Give a suggestion about how to structure workers' compensation so that only the workers who do not intend to retire early choose to go to the training resort.

Exercise 6.5

A profit-maximizing firm has hired a worker for two periods. The worker's productivity is 70 per period. The firm pays wage 70 per period to the worker, which is equal to the worker's outside wage. Firm-specific training could increase the worker's productivity by 50 in period 2. The cost of the training is 30, which includes productivity losses in period 1. The interest rate is zero.

a) Verify that if the worker's wage remains constant, firm-specific training raises the firm's profits.

Suppose the firm has invested in firm-specific training of the worker in period 1.

b) What is the maximum wage the firm is willing to pay to the worker in period 2? (In other words: What is the maximum wage the worker can ask if he can make a take-it-or-leave-it offer to the firm?)

The worker threatens to take another job and leave the firm unless he gets part of the -- in his view exorbitant -- profits in period 2. The firm and the worker settle at an equal split of the surplus in period 2.

c) How much does the worker earn in period 2?

d) What would the firm's decision on firm-specific training be if it had anticipated the worker's threat?

e) How can both the firm and the worker be better off, given that the worker always succeeds in capturing half of the surplus in period 2?

Answers

6.1

a) $50.000 = w + p(-100.000)$, and hence $w = 50.000 + 100.000p$.

b) No. It reduces wage costs from 60.000 to 55.000. But this is not sufficient to make up for the cost (7.500).

c) Wages will be lower: 55.000 rather than 60.000. Since the economy is perfectly competitive, profits were zero, and must become zero again. The government regulation brings a net loss to the firm of 2.500 per worker (see part b)). In the long run, this will imply exit of firms and hence lower supply of goods, and thus higher prices. In an open economy with competition from foreign firms, there may be no effect on prices; Instead, production of these type of goods may completely disappear in this country after introduction of the safety regulation.

d) Worker's lack of information about risks or costs or externalities. As an example of the first, one could assume that workers underestimate the cost of injury (e.g. workers think that $p = 0.10$, but the government knows that p is higher; Information provision by the government may be a better way to deal with the lack of information). An example of the latter is collectively financed medical costs (But a better way to deal with this problem may be to condition medical insurance premiums on occupation).

6.2 *(partly adapted from Peter Kuhn's teaching material)*

a) Profits = $[Q - W] \cdot N(W)$. Differentiate to W and set equal to zero gives:

$$-N(W) + [Q - W] \cdot (dN/dW) = 0$$

Multiply by (w/N) :

$$-W + [Q - W] \cdot (dN/dW)(W/N) = 0$$

Rewrite

$$W = [\eta / (1 + \eta)] Q$$

b) Using the formula $w = \eta Q / (1 + \eta)$, the answers are \$18, \$16, \$12, and \$10. As the labor supply elasticity rises, Alfredsons experiences a greater and greater loss of workers when it cuts wages below \$20. Thus, when $\eta = 9$, Alfredsons experiences a dramatic loss of workers when it cuts wage, so it has no choice but to pay workers almost their full productivity ($Q - w = 20 - 18 = \$2$). When $\eta = 1$, Alfredsons experiences a much smaller loss of workers when it cuts its wage, so its profit-maximizing strategy is to keep half of what each worker produces: ($Q - w = 20 - 10 = \$10$).

6.3

a) Present value of life-time consumption when not studying = $wa + wa/(1+r)$. Present value of life-time consumption when studying = $-t + (wa+pa)/(1+r)$.

People study if: $-t + (wa+pa)/(1+r) > wa + wa/(1+r)$. Rewriting yields: $a > [(1+r)t]/[p - (1+r)w]$.

Differentiate to r gives $da/dr = \{t[p - (1+r)w] + (1+r)tw\}/[p - (1+r)w]^2 > 0$. So, when the interest rate goes up, less people study. Also note that less people study i) when tuition goes up, ii) when the return to education p decreases, iii) when the wage rate w (the reward for innate ability) goes up.

b) The economist's warning can be justified as follows. People study if: $-t + (1-\tau)(wa+pa)/(1+r) > (1-\tau)wa + (1-\tau)wa/(1+r)$. Rewriting yields: $a > [(1+r)t]/\{(1-\tau)[p - (1+r)w]\}$. Clearly, when τ increases, fewer people decide to study. The interpretation of this result follows. A tax on labor income reduces the benefits from education (the returns to education decrease by τpa) as well as the cost (foregone labor income decreases by τwa). Since $pa > wa$, the reduction in the benefits is larger than the reduction in the cost, and so fewer people decide to study.

c) Yes. People study if: $-t(1-\tau) + (1-\tau)(wa+pa)/(1+r) > (1-\tau)wa + (1-\tau)wa/(1+r)$. Rewriting yields: $a > [(1+r)t]/\{[p - (1+r)w]\}$. The tax rate no longer plays a role in study choices. Try to figure out why this is the case!

6.4

a) Only those workers for whom the following condition holds:

$$\sum_{t=1}^T \frac{10}{(1+r)^t} > 50 + \frac{1}{4}100 = 75.$$

The returns to training increase with T . Relatively young workers (those with a high T) should be sent to the training.

$$b) (1-\pi) \sum_{t=1}^T \frac{10}{(1+r)^t} + \pi \sum_{t=1}^{T-10} \frac{10}{(1+r)^t} > 75.$$

The possibility of early retirement reduces the returns to training. The value of T above which training is optimal increases.

c) One suggestion is to promise a higher wage for trained workers in the years $T-10$ til T accompanied by a lower wage in the years before $T-10$. If the difference in wages is big enough, only those workers who know they will not retire after $T-10$ years will choose to be trained.

6.5

a) Productivity increases by 50, training cost is 30, so profits increase by 20.

b) 120.

c) Surplus = $120 - 70 = 50$. So worker's wage is $70 + \frac{1}{2} 50 = 95$.

d) Given the wage in period 1, it would have decided not to train the worker. Without firm-specific training, total profits are 0. With firm-specific investment, total profits are $70 - 70 - 30 + 120 - 95 = -5$.

e) By sharing the cost of training. Worker earns 55 in period 1, 95 in period 2. This is more than 2×70 elsewhere or without training. Firm earns $70 - 55 - 30 = -15$ in period 1 and $120 - 95 = 25$ in period 2. In sum (+10), this is more than without training the worker (0).